

Improving Cross-lingual Aspect-based Sentiment Analysis with Sememe Bridge

YIJIANG LIU, Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, China and Huawei Terminal Cloud Services Xiaoyi Business Department, Shenzhen, China

FEI LI, Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, China

DONGHONG JI, Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, China

Aspect-based Sentiment Analysis (ABSA) comprises numerous subtasks including aspect term extraction (AE), opinion term extraction (OE), opinion pair extraction (PE), and triplet extraction (TE). Current research in Chinese ABSA primarily concentrates on aspect terms and sentiment polarity, with insufficient emphasis on opinion terms. This article aims to provide a viable solution for the unannotated Chinese ABSA subtasks such as OE, PE, and TE. First, we develop an English-Chinese parallel dataset for ABSA using a semi-automatic process involving machine translation and a word aligner. Second, we examine the efficacy of cross-lingual transfer methods. Third, we propose a plug-and-play transfer method based on sememe knowledge. Sememes are the language-independent smallest semantic units that encapsulate the components, commonalities, and attributes of things extracted from the real world, which can bridge the gap between English and Chinese. Experimental results show that our proposed method brings significant improvements for Chinese ABSA, and achieves a maximum increase of 8% on the F1 metric for model transfer and label transfer on OE, PE, and TE.

CCS Concepts: • **Computing methodologies** → **Information extraction**; **Language resources**;

Additional Key Words and Phrases: Aspect-based sentiment analysis, cross-lingual learning, zero-shot transfer learning, sememe knowledge

ACM Reference Format:

Yijiang Liu, Fei Li, and Donghong Ji. 2024. Improving Cross-lingual Aspect-based Sentiment Analysis with Sememe Bridge. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 23, 12, Article 166 (November 2024), 22 pages. <https://doi.org/10.1145/3691342>

This work is supported by the National Natural Science Foundation of China (No. 61772378), the National Key Research and Development Program of China (No. 2017YFC1200500), the Research Foundation of Ministry of Education of China (No. 18JZD015), the Major Projects of the National Social Science Foundation of China (No. 11&ZD189).

Authors' Contact Information: Yijiang Liu, Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, Hubei, China and Huawei Terminal Cloud Services Xiaoyi Business Department, Shenzhen, China; e-mail: cslyj@whu.edu.cn; Fei Li, Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, Hubei, China; e-mail: foxlf823@gmail.com; Donghong Ji (Corresponding author), Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan, Hubei, China; e-mail: dhji@whu.edu.cn.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](https://permissions.acm.org).

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 2375-4699/2024/11-ART166

<https://doi.org/10.1145/3691342>

1 Introduction

User-generated reviews express their opinions on used products in multiple aspects, which are valuable to potential consumers and product manufacturers. **Aspect-Based Sentiment Analysis (ABSA)** [46] is a fine-grained sentiment analysis task, which has attracted abundant attention in recent years [40, 44, 45]. ABSA has derived a number of subtasks, which all revolve around predicting three major elements of ABSA, i.e., aspect term, opinion term and sentiment polarity, or their combinations. For example, **aspect term extraction (AE)** [36, 68] and **opinion term extraction (OE)** [13, 59] seek to extract all aspect and opinion terms. Aspect-opinion **pair extraction (PE)** [62, 72] aims to extract the aspect terms and the corresponding opinion terms simultaneously. **Triplet extraction (TE)** [11, 69] targets extracting all pairs and their sentiment polarity. We illustrate the definitions of these subtasks in Figure 1 with specific examples.

Benefiting from the rich resource of benchmark datasets, English ABSA has made great progress. Most works of Chinese ABSA focus on aspect term extraction and sentiment classification [6, 32, 67, 73]. However, none of these works focus on opinion terms, let alone aspect-opinion pairs or triplets, which affects the integrity of Chinese ABSA systems.

Cross-lingual learning is a possible solution to the problem of low resources [18, 57], especially to transfer the advantage of the rich source language (such as English) resources to the target language with scarce or even no annotated data. There are two representative methods of zero-shot cross-lingual transfer learning: **model transfer** and **label transfer** [50]. As shown in Figure 2, model transfer trains the model using the original English training set, while label transfer uses the automatically obtained Chinese dataset. Both of them perform model evaluation on the manually checked Chinese test set.

This work focuses on Chinese ABSA. In particular, we carry out the first study for Chinese OE, PE and TE. First, we construct a parallel dataset between English and Chinese for ABSA through machine translation, word alignment and manual correction following previous works [19, 75]. Second, we conduct cross-lingual learning through unsupervised model transfer and label transfer, setting up strong baselines for future research. Third, we propose a plug-and-play transfer method based on sememe knowledge provided by the bilingual sememe knowledge base [15]. As shown in Figure 1, sememe is a language-independent smallest semantic unit [5], which can be used as an intermediate medium to transfer the task-specific knowledge and bridge the gap between source and target languages. Sememes are the smallest semantic units that encapsulate the components, commonalities, and attributes of things extracted from the real world. Aspect words and opinion words often share similar characteristics, hence their sememe compositions are also similar. Therefore, the sememes trained in the source language can provide strong features for inference in the target language. This sememe transfer method acts as a stand-alone sememe encoding module that can be accessed directly without changing the underlying model structure.

Extensive experiments are conducted on the newly constructed datasets. The results show that transfer learning achieved decent results for Chinese ABSA on the newly constructed dataset. Meanwhile, our proposed plug-and-play transfer method has significant effects on both model transfer and label transfer, with a maximum increase of 8% on the F1 metric for OE, PE, and TE.

In summary, our contributions are three-fold:

- We explored Chinese OE, PE, and TE for the first time and constructed a parallel dataset of Chinese and English. We investigate the performance of transfer learning on the newly constructed dataset, establishing a strong baseline for future works.
- We propose a plug-and-play transfer method, which uses sememe as an intermediate medium to transfer the knowledge learned from annotated data in the source language.

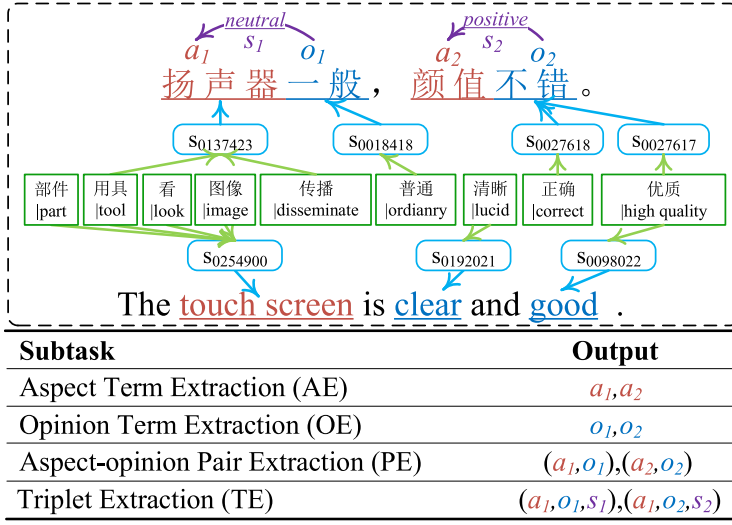


Fig. 1. Definitions of ABSA subtasks and architecture among sememe (green), sense (blue), and word in English and Chinese. S_x indicates the number of the sense in HowNet. A sememe may be shared across senses, e.g., the ‘用具|tool’. Note that the sentences here are not parallel, which is consistent with the actual process of training and inference.

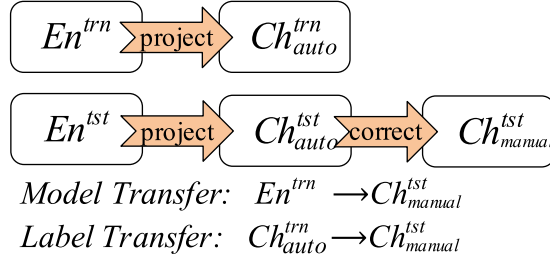


Fig. 2. Illustration of cross-lingual learning methods. ‘trn’ and ‘tst’ mean training set and test set, respectively. ‘ $A \rightarrow B$ ’ means using A as the training set and B as the test set.

- Experimental results show that the semi-automatic data construction method and transfer learning is a possible solution for Chinese ABSA, and our proposed method brings significant improvements on both model transfer and label transfer.

2 Related Work

2.1 Aspect-based Sentiment Analysis

Pontiki et al. [46] organized the first SemEval competition, which is a milestone for Aspect-Based Sentiment Analysis (ABSA). Then ABSA became one of the research hotspots for NLP, followed by many related studies. **Aspect Term Sentiment Analysis (ATSA)** [55] aims to analyze the sentiment polarity of a given aspect term that appears in the text. **Aspect Term Extraction (AE)** [36, 68] aims to extract explicit aspect expressions from text. **Aspect-sentiment Pair Extraction (APE)** [27] unites ATE and ATSA to obtain aspect terms and sentiment polarities simultaneously. **Aspect-Opinion Pair Extraction (PE)** [10, 72] is to jointly extract aspect terms and opinion terms. **Aspect-opinion-sentiment Triplet Extraction (TE)** [40, 69] takes aspect term, opinion term, and

sentiment polarity extraction together into consideration, and provides a unified fine-grained solution for ABSA.

Chinese ABSA mainly focus on aspect terms and sentiment polarity. Liu [30] draws a relatively complete picture for opinion, which is defined as a quintuple $\langle \text{target entity, aspect, sentiment value, holder, time} \rangle$. However, most of the follow-up Chinese-oriented works mainly focus on ATSA [9, 29], only predicting the sentiment polarity of a given aspect in the text. Some Chinese ABSA works have been expanded to some extent. Yang et al. [67] propose a multi-task learning model for Chinese-oriented aspect-sentiment pair extraction, which equips the capability of extracting aspect term and inferring aspect term polarity synchronously. Bu et al. [6] jointly model aspect category sentiment analysis and review rating prediction, providing a large-scale Chinese dataset. However, the opinion term-related subtasks under Chinese ABSA is still an untrodden area.

2.2 Cross-lingual Learning

Cross-lingual learning is a possible way in many tasks for low resource languages [71], which attempts to use the rich resource of annotated datasets and break the limitation caused by insufficient training data. Cross-lingual learning has been proved effective in many NLP tasks, including sentiment analysis [2, 60], named entity recognition [4, 43], semantic role labeling [19, 20], dependency parsing [1, 8], and opinion role labeling [75]. The commonly used corpus include Dependencies [38], CoNLL [51, 52], Amazon reviews [47], XNLI [47], DARPA LORELEI [54], Wikipedia,¹ and Reuters Collections [3].

The two most common seen transfer methods are model transfer and label transfer [41]. **Multi-lingual BERT (MBERT)** and **Adapter** are the two most used model transfer methods. MBERT [14] embeds words from multiple languages into one semantic vector space, in which semantically similar words are close together independently of the language. Adapter [22] adds two non-linear layers into the transformers of MBERT to train the entire pre-trained model by optimizing a small number of parameters instead. Model transfer has limited representation abilities for languages of different language families, which usually limits the performance of downstream cross-lingual tasks. Label transfer is model independent, which establishes a correspondence between the source and target languages. Label transfer transfers labels between the samples from different languages, thus the projected labels can then be used for further training. The main steps of the label transfer consist of sample correspondence and label projection [33, 41]. Sample correspondence mainly uses parallel corpus or translation systems, while the translation process may introduce noise. Projection of complex labels may also lead to some mislabeling.

2.3 Sememe and HowNet

One word may have multiple meanings, and even a meaning can be further divided into smaller units. For example, the common meaning of the English word “dog” can be expressed by the combination of the meanings of “livestock” and “domesticated”. The minimum indivisible semantic units are defined as sememes [5]. HowNet is a famous sememe knowledge base, which defines the objective world in terms of components, attributes, time, space, and events [15]. Through the bottom-up induction method, a limited set of sememes are determined. HowNet takes words (for both Chinese and English) as a unit and annotates each senses of words with sememes and inter-sememe relationships.

Sememe and HowNet have been introduced into many NLP tasks. Duan et al. [16] resolve lexical ambiguity through sememes. Niu et al. [37] propose a sememe encoding module for enhanced word representation learning. Liu et al. [31] combine the sentiment dictionary and sememes of

¹<https://wikipedia.org/>

specific fields to classify the polarity of Chinese online product reviews. Xu et al. [66] annotate the subjectivity and sentiment polarity of terms in HowNet and utilizes them to construct an opinion mining system. Luo et al. [34] utilize the topology between sememes to enhance the word representation for ABSA. Qi et al. [48] investigate how to automatically build a sememe knowledge base in a new language.

Overall, the English ABSA task is more active in the NLP community with a large amount of annotated corpus and has grown significantly due to its generality. The lack of Chinese corpus annotated with opinion terms limits the further exploitation of Chinese ABSA systems. To address the problem and advance related researches, this work focus on Chinese ABSA, especially opinion term-related subtasks, i.e., OE, PE, and TE. We manually constructed an English-Chinese parallel corpus and explored the effectiveness of cross-lingual learning on Chinese ABSA. Further, we propose a plug-and-play transfer module using a language-independent sememe knowledge, which provides a new way of knowledge transfer in addition to model transfer and label transfer, the two commonly used transfer methods.

3 Task and Dataset

3.1 Task Definition

We now introduce the definition of the subtasks of ABSA in Figure 1. For a given review consisting of a sequence of words $x = [x_1, x_2, \dots, x_n]$, the goal of AE and OE is to decode all aspect terms $\{a_i\}_{i=1}^{N_a}$ and opinion terms $\{o_i\}_{i=1}^{N_o}$, respectively. Both a_i and o_i are composed of continuous tokens of x . The goal of PE is to decode all $\{(a, o)_i\}_{i=1}^{N_p}$ pairs. As for TE, it additionally extracts the corresponding sentiment polarity of each opinion pair, i.e., extracting all $\{(a, o, s)_i\}_{i=1}^{N_t}$ triplets, where $s \in \{positive, neutral, negative\}$.

Under the settings of zero-shot cross-lingual ABSA, the training set and valid set are in the source language ℓ_s , and the test set is in the target language ℓ_t .

3.2 Dataset Construction

We induce Chinese ABSA data from the annotated English data automatically by translating the source sentences and then project the aspect and opinion terms into their corresponding words by aligning. Thus, the pairs and triplets can be projected by keeping the original relationships. This method has been demonstrated effective for cross-lingual dependency parsing [56, 70], semantic role labeling [19], and opinion role labeling [75]. In addition, only aspect and opinion terms are required to be projected into the target side rather than all sentential words, which reduces the error during the process. The whole process consists of three steps: machine translation, label projection, and manual correction.

Machine Translation. **Neural machine translation (NMT)** has made impressive advances in recent years [25, 64], which enables our method to have high-quality translations. We use a state-of-the-art translation system² to produce the target translations for the sentences of the source ABSA data.

Label Projection. Then we incrementally project the corresponding aspect terms or opinion terms of a source sentence to its target. Before the projection, we use the segment tool [35] for Chinese word segmentation. The projection is supported by word alignments, which can be produced with the word-alignment tool [17].

²<https://fanyi.baidu.com>

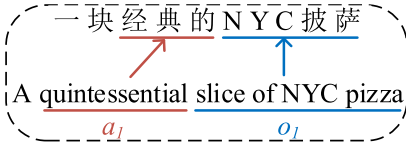


Fig. 3. Example of label projection from English to Chinese.

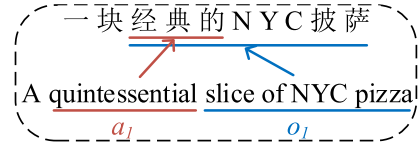


Fig. 4. Example of overlapped projected labels from English to Chinese.

Table 1. Errors in Several Automatically Obtained Samples and Results of Manual Correction

Translation Error	Original	Our server was very professional and friendly .
	Auto	我们的[服务器] _{AT} 非常[专业] _{OT} 、[友好] _{OT} 。
	Revision	我们的[服务员] _{AT} 非常[专业] _{OT} 、[友好] _{OT} 。
Poor Translation	Original	Dessert can't be missed , so save room!
	Auto	[甜点] _{AT} [不可错过] _{OT} , 所以节省空间!
	Revision	[甜点] _{AT} [不可错过] _{OT} , 所以留点肚子!
Alignment Error	Original	The food was boring and expensive .
	Auto	[食物] _{AT} [既单调] _{OT} 又[昂贵] _{OT} 。
	Revision	[食物] _{AT} 既[单调] _{OT} 又[昂贵] _{OT} 。

Words marked in red and blue are aspects and opinions, respectively.

Given an English sentence $\{e_1 \cdots e_n\}$ and a Chinese sentence $\{c_1 \cdots c_n\}$, the alignment tool establishes a mapping between these two sentences to form a bipartite graph, in which the connected nodes forms an aligned word pair. As shown in Figure 3, for each aspect or opinion term $e_{i,j}$ in the English sentence, the corresponding Chinese span $c_{i',j'}$ are assigned the same label (aspect or opinion). The pairs and triplets are established automatically once the terms have been projected.

As shown in Figure 4, there is one case that requires special treatment: overlapping, which means that projected terms in the target sentence are overlapped. In this case, we regard the overlapping words as a part of the shorter or the former ones.

Manual Correction. The last step is to perform manual checking to ensure the quality of the automatic translation and projection. There are several problems that need to be corrected, including translation errors, poor translation, and boundary alignment errors as shown in Table 1.

We let three native Chinese speakers revise the errors in translation, correct the labels obtained by automatic alignment, and add the missing annotations. All three annotators are pursuing a master's degree in computer science with considerable familiarity with ABSA tasks. They possess a strong command of the English language, having passed the CET-6 test, which can be approximately equated to score 6 in IELTS. Their proficiency ensures the annotations are consistent with the nuanced meanings expressed in both English and Chinese contexts. Taking the source sentence and the labels as a reference, there is almost no ambiguity in the manual correction of translation and projection. During the annotation phase, the first two annotators shall work on each data point to ensure the reliability of the annotations. When there is ambiguity, the third annotator makes the final choice. Finally, Cohen's Kappa score of Aspect-opinion-sentiment Triplet is 0.90, which indicates our annotation corpus has reached a high-level agreement.

4 Method

We take the advanced span-based model as our backbone, which achieves advanced performance in many information extraction tasks [65, 72]. Figure 5 shows the overall architecture of the model

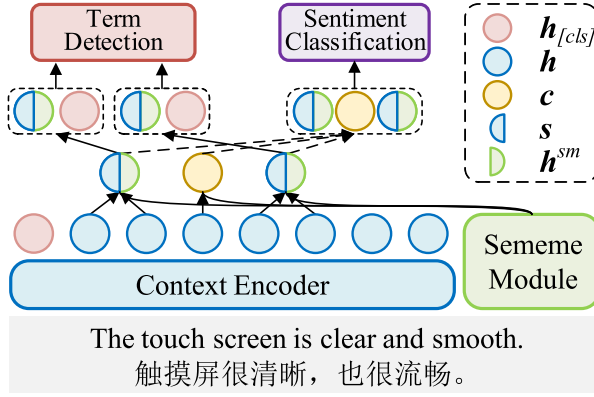


Fig. 5. Overall architecture of the backbone model.

we used. Based on the basic model, we add a sememe encoder as a bridge for knowledge transfer between English and Chinese.

4.1 Basic Model

Encoder. We exploit the **Multi-lingual BERT (MBERT)** [14] as the basic encoder because it has powerful cross-language representation capabilities [23, 42] and has achieved state-of-the-art performances for a number of cross-lingual tasks [19, 75]. For a given input $x_1 \cdots x_n$, the output of MBERT is a sequence of contextual representation at the token-level:

$$h_{[cls]}, h_1, \dots, h_n, h_{[sep]} = \text{BERT}(x_1, \dots, x_n), \quad (1)$$

where n is the length of input, h is our desired representation.

Span Representation. After encoding the whole sentence, we then enumerate all possible spans to generate the candidates for aspect or opinion terms. Each span $s_i = \{x_{t(i)}, \dots, x_{e(i)}\}$ is defined as a single word or phrase with a start index $t(i)$ and an ending index $e(i)$. For example, when setting the maximum length of span to 3, the token sequence ‘the eating place is ...’ will be mapped into text spans: ‘the’, ‘eating’, ‘place’, ‘the eating’, ‘eating place’, ‘the eating place’, ...

Based on the token representation h , we construct the span term representation s :

$$\begin{aligned} s'_i &= [h_{head}; h_{tail}; h_{size}; h_{pool}], \\ s_i &= \text{FFN}_1(\text{Dropout}(s'_i)) \end{aligned} \quad (2)$$

where FFN_1 is a feed-forward layer, dropout module [53] is used for relieving the overfitting of span representations. h_{head} and h_{tail} are the boundary representation of the start and the end of the span, respectively. h_{size} is the span width embedding which is found by using a look-up table. h_{pool} is the pooling representation over the current span, which is obtained via Avg-Pooling operation:

$$h_{pool} = \text{Avg-Pooling}(h_{head}, \dots, h_{tail}). \quad (3)$$

As the model enumerates all possible span terms, the time complexity is proportional to the number of spans given a sentence of n words, i.e., $O(n^2)$. We prune the span numbers by limiting the maximum length of each span with a fixed number Φ .

Term Detection. We next determine the span type based on the span representation $\mathbf{s}$ and the context representation $\mathbf{h}_{[cls]}$ yielded from MBERT. We employ a softmax layer on these term representations to obtain the term types:

$$\mathbf{p}_i^d = \text{Softmax}(\text{FFN}_2([\mathbf{s}_i; \mathbf{h}_{[cls]}])) \quad (4)$$

where $\mathbf{p}_i^d$ is the probability of term type $t \in T = \{aspect, opinion, none\}$, which means aspect term, opinion term, and neither, respectively.

Sentiment Classification. For a term pair $g_{i,j}$ composed of $\langle s_i, s_j \rangle$, we obtain the term pair representation $\mathbf{g}_{i,j}$ by coupling both term representations and their local context $\mathbf{c}_{i,j}$:

$$\begin{aligned} \mathbf{g}_{i,j} &= \text{FFN}_3([\mathbf{s}_i; \mathbf{c}_{i,j}; \mathbf{s}_j]), \\ \mathbf{c}_{i,j} &= \text{Avg-Pooling}(\mathbf{h}_{e(i)+1}, \dots, \mathbf{h}_{t(j)-1}) \end{aligned} \quad (5)$$

where local context $\mathbf{c}_{i,j}$ is obtained via Avg-Pooling operation on the tokens between two terms.

We next input the term pair representation to a softmax layer to obtain the sentiment polarity:

$$\mathbf{p}_{i,j}^p = \text{Softmax}(\mathbf{g}_{i,j}) \quad (6)$$

where $\mathbf{p}_{i,j}^p$ is the probability of sentiment polarity type $p \in \{positive, neutral, negative, none\}$, which means the sentiment polarity of (s_i, s_j) is *positive*, *neutral*, *negative*, or (s_i, s_j) cannot form an opinion pair, respectively.

For the PE task, we ignore the category of sentiment polarity and regard it as an aspect-opinion pair as long as the label is not *none*.

Training. The training objective is defined as the sum of the negative log-likelihood from both the term detection and term pairing:

$$\begin{aligned} \mathcal{L}^d &= - \sum_{s_i \in S} \log \mathbf{p}^d(\hat{t}_i | s_i), \\ \mathcal{L}^p &= - \sum_{g_{i,j} \in G} \log \mathbf{p}^p(\hat{r}_{i,j} | g_{i,j}), \\ \mathcal{L} &= \mathcal{L}^d + \alpha \mathcal{L}^p \end{aligned} \quad (7)$$

where $\hat{t}_i$ is the gold term type of span s_i , $\hat{r}_{i,j}$ is the gold sentiment polarity of span pair $g_{i,j}$. S and G indicate the enumerated span and triplet pool, respectively. α is a hyper-parameter to balance these two loss functions.

4.2 Sememe Module

4.2.1 Sememes in HowNet. Sememe is a language-independent smallest semantic unit [5], which has been exploited for several NLP tasks, such as word sense disambiguation [21], event detection [26], and relation extraction [28]. HowNet is one of the most famous sememe knowledge bases, which defines more than 100,000 words in both Chinese and English with 2,187 sememes [49].

To elucidate the connective role of sememes across two languages, our exploration commences with parallel samples, and thereafter expands to more general instances. Figure 6 illustrates that both aspect and opinion terms contained within parallel sentences can align with an entirely congruent set of sememe combinations. Therefore, in an extreme case where the context representation of words is not considered (such as the contextual representation obtained from Multilingual BERT for “touch screen” and “触摸屏”), when we obtain well-trained representations of sememes on the English training set (e.g., “部件|part,” “用具|tool,” “看|look,” “图像|image”), we can achieve

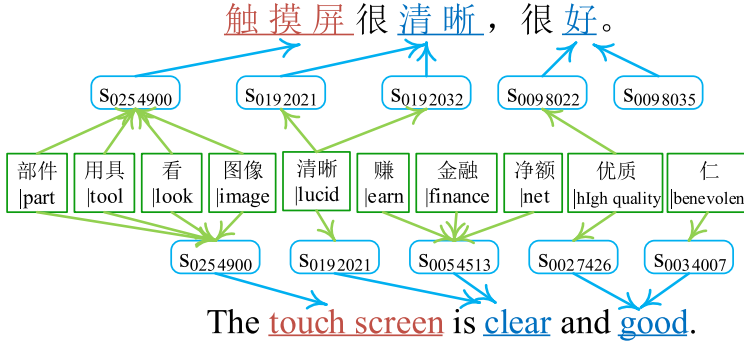


Fig. 6. Illustration of the connective role of sememes in parallel samples. Blue squares represent the senses of Chinese and English words in HowNet, where a word can have multiple senses. Green squares represent the sememes corresponding to these senses in HowNet.

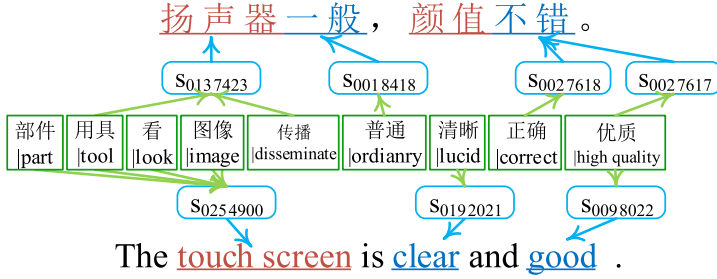


Fig. 7. Illustration of connective role of sememes in non-parallel samples. Blue and green squares represent the senses and sememes in HowNet, respectively.

robust confidence in predicting “触摸屏” as an aspect term by locating its corresponding sememes through HowNet.

We now proceed to a more general example: instances where Chinese and English are non-parallel. As illustrated in Figure 7, it can be seen that even when the sentences are not parallel, connections between Chinese and English words can still be established via sememes. However, it should be noted that these connections may not be as robust as those found in parallel instances. For instance, the sememe “用具|tool” is shared between “touch screen” and “扬声器”. In the process of training, sememes decompose the span-level features learned in the English corpus and recombine them during inference in Chinese, so as to achieve the effect of cross-lingual knowledge transfer.

4.2.2 Sememe to Span Representation. Adding sememe knowledge at the span level involves two steps, matching and fusion. We first find the matching words and corresponding sememes according to the text of span and then fuse the sememe information.

Matching. Since the text of span may or may not exist in the HowNet, we use the set of words defined in HowNet as the dictionary to segment span in the way of forward maximum matching. All matched senses and sememes are regarded as the component of the span. For each span $s \in S$, there are multiple senses $sn_i^{(s)} \in SN^{(s)}$, where $SN^{(s)}$ denotes the senses of s . Each sense $sn_i^{(s)}$ consists of several sememes $sm_j^{(sn_i)} \in SM^{(sn_i)}$, where $SM^{(sn_i)}$ represents the sememes of $sn_i^{(s)}$.

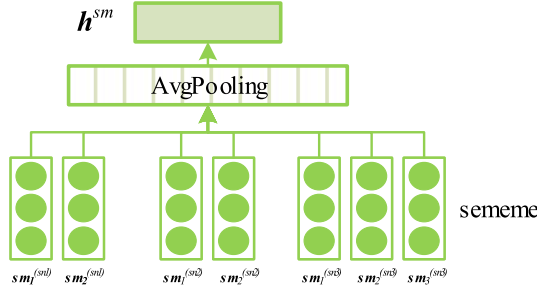


Fig. 8. Illustration of SF.

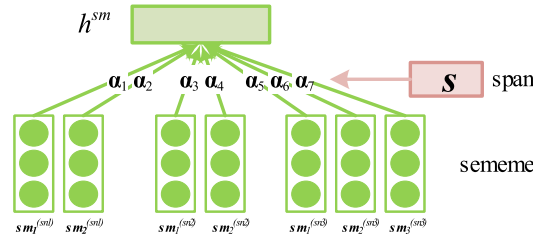


Fig. 9. Illustration of SAF.

Fusion. we project each sememe $sm_j^{(sn_i)}$ to a vector sm_i through an embedding lookup table $W_{sm} \in \mathbb{R}^{d_{sm} \times |sm|}$, where d_{sm} is the dimension of sememe features, and $|sm|$ indicates the size of the sememe set. For a span s , we propose the following three sememe-encoded models to inject sememe information:

- (1) **Simple Flat (SF).** As illustrated in Figure 8, SF treats the corresponding sememes $sm_i^{(s)} \in SM^{(s)}$ of span s all the same, and uses the average of all its sememes as the sememe-encoded representation:

$$h^{sm} = \frac{1}{|SM^{(s)}|} \sum_{sm_j \in SM^{(s)}} sm_j^{(s)} \quad (8)$$

where $|SM^{(s)}|$ denotes sememe number of span s .

- (2) **Sememe Attention over Flat (SAF).** As illustrated in Figure 9, SAF treats sememes in a flat structure like SF, and calculates the weights of sememes $sm_i^{(s)} \in SM^{(s)}$ of span s using the attention mechanism. The sememe-encoded representation is obtained by weighted summation:

$$\begin{aligned} h^{sm} &= \sum_{sm_i \in SM^{(s)}} \alpha_i^{sm} W_1 sm_i, \\ \alpha_i^{sm} &= \frac{\exp(e_i)}{\sum_{sm_j \in SM^{(s)}} \exp(e_j)}, \\ e_i &= \sigma(\text{FFN}_4[W_2 s \| W_2 sm_i]) \end{aligned} \quad (9)$$

- (3) **Sememe Attention over Tree (SAT).** As illustrated in Figure 10, SAT organizes sememes in senses, and calculates the weights of senses $sn_i^{(s)} \in SM^{(s)}$ about the span. Sense representation is obtained by averaging all the corresponding sememe $sm_i \in SM^{(sn_i)}$ of

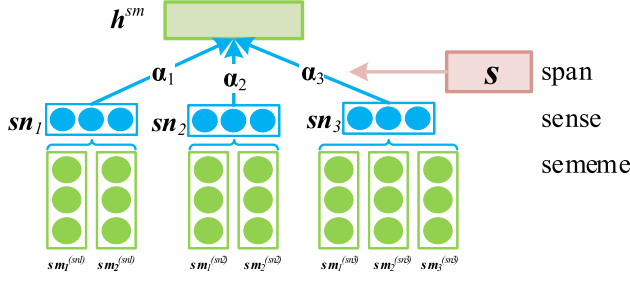


Fig. 10. Illustration of SAT.

sense sn_i :

$$\begin{aligned}
 h^{sm} &= \sum_{sn_i \in SN(s)} \alpha_i^{sn} W_1 sn_i, \\
 \alpha_i^{sn} &= \frac{\exp(e_i)}{\sum_{sn_j \in SN(s)} \exp(e_j)}, \\
 e_i &= \sigma(\text{FFN}_4 [W_2 s \| W_2 sn_i]), \\
 sn_i &= \frac{1}{|SM^{(sn_i)}|} \sum_{sm_j \in SM^{(sn_i)}} sm_j
 \end{aligned} \tag{10}$$

where $|SM^{(sn_i)}|$ denotes the sememe number of sense sn_i .

Finally, we concatenate the span representation s and the corresponding sememe representation h^{sm} to replace s in Equation (2) as the input of subsequent modules: $s = [s; h^{sm}]$.

- (4) Comparison. Each of these methods presented for integrating sememes involves two key steps: matching sememes to words within a span of text and then fusing this sememe information into the word's representation.

SF does not differentiate the importance of different sememes; each contributes equally to the final representation. This method is straightforward but may lose nuances by not weighting sememes based on relevance or context.

The key distinction between SAF and SF is the attention mechanism, which allows the model to focus more on certain sememes that are deemed more relevant to the text span. This enables more context-sensitive sememe representations by acknowledging that some sememes contribute more to the meaning of the span than others.

Unlike SF and SAF, SAT takes into account the multi-sense structure of language, where words can have multiple meanings. By first averaging sememes within a sense and then applying attention at the sense level, SAT is able to capture which set of sememes is most applicable to the text span in context. This potentially captures more accurate semantic information by considering the word's intended meaning in the specific context.

These methods exhibit significant differences in the following dimensions:

- Context Sensitivity: SF has the lowest context sensitivity as it does not differentiate between sememes. SAF and SAT are better at capturing context, with SAT likely being the most effective due to its hierarchical attention mechanism.
- Computational Complexity: SF is the simplest and thus the least computationally demanding. SAF introduces additional computation through attention weights, and SAT is even more complex due to its two-level attention mechanism.

Table 2. Data Statistics of Our Corpus, where #S, #A, #O, and #T Refer to the Number of Sentences, Aspect Terms, Opinion Terms and Triplets, Respectively

Lan.	Dataset	Res				Lap			
		#S	#A	#O	#T	#S	#A	#O	#T
En	Train	2,313	3,943	3,063	3,395	2,934	2,818	2,845	2,346
	Valid	602	982	736	823	326	308	318	259
	Test	937	1,652	1,292	1,458	816	804	775	673
Ch_{auto}	Train	2,313	3,824	2,933	3,188	2,934	2,767	2,731	2,214
	Valid	602	949	715	774	326	300	298	236
Ch_{manual}	Train	2,313	3,889	2,956	3,240	2,934	2,845	2,806	2,298
	Valid	602	962	718	800	326	306	310	252
	Test	937	1,681	1,325	1,486	816	811	766	661

The number of pairs is the same as #T.

- Semantic Precision: SAT is likely to offer the highest semantic precision because it accounts for different word senses and applies attention to determine the most contextually relevant sememes.
- Model Generalization: Models with attention mechanisms like SAF and SAT might generalize better to different contexts by focusing on the most relevant sememes in various situations, as opposed to SF, which treats all sememes the same regardless of the context.

In conclusion, while all three methods provide ways to incorporate sememes into span representations, they differ significantly in their strategies for weighting and organizing sememe information. SF offers simplicity and speed, while SAF and SAT aim to capture more nuanced semantic information through the use of attention mechanisms, with SAT providing additional sophistication by considering word senses.

5 Experiments

5.1 Evaluation

We calculate exact precision (P.), exact recall (R.) and **exact F1** values for PE and TE. An opinion pair or triplet is regarded as correct if all the elements are correctly recognized in terms of boundaries and categories. Besides **exact F1**, we take the **binary F1** and **proportional F1** scores as the extra evaluating metrics for AE and OE following Xia et al. [63]. All results of our models are presented after a significant test with the best baselines, with $p \leq 0.05$.

5.2 Dataset

To evaluate our models, we construct a parallel dataset of English and Chinese involving the field of restaurant and laptop. We combine three English restaurant datasets released by Wang et al. [58], and remove the duplicate samples. The English laptop dataset we used is published by Cai et al. [7]. Each dataset is divided into fixed training, validation, and test sets. We construct the Chinese dataset following Section 3.2.

First, we obtain the training, validation, and test sets automatically by machine translation and alignment. Then, the test set is obtained after manual correction with 90% consensus. Table 2 shows the basic statistics of the corpus.

To evaluate the quality of the automatically obtained Chinese training set, we calculate the evaluation metric between Ch_{auto}^{trn} and Ch_{manual}^{trn} , by setting Ch_{manual}^{trn} as the ground truth and Ch_{auto}^{trn} as the prediction. Note that the absolute position of the term here may deviate due to

Table 3. Evaluation Metrics of the Automatically Obtained Chinese Training Set Ch_{auto}^{trn} Compared with the Manually Corrected Training Set Ch_{manual}^{trn}

	P	R	F1
Exact metric			
Aspect	71.37	71.57	71.47
Opinion	61.43	61.59	61.51
Pair/Triplet	47.94	47.73	47.83
Proportional metric			
Aspect	79.18	80.53	79.85
Opinion	66.68	66.85	66.76
Binary metric			
Aspect	83.42	84.85	84.12
Opinion	79.54	79.75	79.65

manual correction of the sentence translation, so it is considered correct here as long as the term is consistent.

Table 3 shows the results of the evaluation metrics. We found that decent corpus could be obtained through machine translation and label projection. We also found that: (1) The F1 values of aspect term is higher than that of opinion term, which is due to the fact that the expression of opinion is more diverse and brings a challenge to the alignment process. (2) The binary F1 is much higher than the exact F1, indicating that in many samples the boundary identification is not completely correct.

5.3 Models

We use model transfer and label transfer as benchmark models. Model transfer trains the model using the original English training set, while label transfer use the automatically projected Chinese dataset. Both of them perform evaluation on the Chinese test set. Multi-lingual BERT (MBERT) and Adapter are the two most common seen model transfer methods.

MBERT. We utilize Multi-lingual BERT³ as the context encoder, which can embed semantically similar words across different languages into closer representations. En^{trn} is used as the training set.

Adapter. Adapter [22] adds two non-linear modules into the transformers and trains the entire model by optimizing a small number of parameters. En^{trn} is used as the training set.

Alignment-based Label Transfer (A-LaTran.) We use the data obtained by automatic translation and alignment (without manual correction) as the training set. Chinese BERT⁴ is utilized as the context encoder. Ch_{auto}^{trn} is used as the training set.

Marker-based Label Transfer (M-LaTran.) Marker-based Label Transfer [12] uses the “Mark-then-translate” approach, where special markers are first inserted around the annotated spans in the text of the source language. The modified sentence with markers inserted is then fed as input to an machine translation system, projecting the label span markers to the target sentence as a byproduct of translation.

³<https://huggingface.co/bert-base-multilingual-uncased>

⁴<https://huggingface.co/bert-base-chinese>

ChatGPT. Large language models have shown powerful capabilities in zero-shot, few-shot, cross-lingual, and cross-domain tasks [24, 74]. With the emergence of instruction learning [39], Large interactive language models have attracted widespread attention. Dialogue products based on these models, such as ChatGPT and NewBing, have caused quite a sensation in both the academic and industrial communities. This paper uses the gpt-3.5-turbo-0301 model provided by ChatGPT through OpenAI's API,⁵ and experiments under the zero-shot setting. We manually design five prompts, and report the best performing results.

MualTrn. In order to explore the performance difference between unsupervised cross-lingual learning and supervised learning, and understand how much drop is occurring caused by the automatically constructed training set holistically, we added one more setting to use the manually corrected training set Ch_{manual}^{trn} for training without sememe module. This setting can reflect the upper limit of the performance of the other models to some extent.

Note that, for comparing the performance differences between the model transfer and label transfer, and quantifying the improvements our methods can achieve in both cross-lingual learning paradigms, we use Multilingual BERT for model transfer and Chinese BERT for label transfer, respectively.

5.4 Implementation Details

We take the pre-trained language model MBERT released by Hugging Face [61] as the context encoder in our model. We adopt the best values according to the results on the valid dataset for the hyperparameters. We adopt the dropout mechanism for the MBERT output representation with 0.3 dropping rate. The dimension of contextualized token representation $\mathbf{h}$ is 768. The sememe dimension d_{sm} is set to 64. The output dimensions of FFN₁ and FFN₃ are set to the same as $\mathbf{h}$. We set the adapter size to 64 for Adapter. The maximum length of each span Φ is set to 6. We optimize all model parameters (including the BERT part) by standard back-propagation. The learning rate is set to $5.0e-5$ and $1.0e-3$ for the context encoder (MBERT and Adapter) and the remaining parameters, respectively. The training epoch is set to 100 with a 10% training steps linear warmup. We do an early stop if results on the valid set do not improve for 30 epochs. All experiments are conducted with an NVIDIA GeForce GTX 3090 GPU and 24 GB graphic memory.

6 Results and Analyses

We try to answer several questions via the studies in this section. **Q1** Is transfer learning a possible solution for Chinese ABSA on the newly constructed dataset? **Q2** Is sememe knowledge helpful for both model transfer and label transfer? **Q3** Which sememe fusion method is the most effective? **Q4** Is it reasonable to utilize sememes for cross-lingual transfer? **Q5** How effective is the proposed approach in more challenging tasks, such as cross-lingual and cross-domain ABSA? For Q1, refer to Sections 6.1 and 6.2. For Q2 and Q3, refer to Section 6.3. For Q4 and Q5, refer to Sections 6.5 and 6.6, respectively.

6.1 Overall Results

Tables 4 and 5 show the experimental results of PE, TE, AE, and OE, respectively. Overall, transfer learning achieved decent results under zero-shot setting on the newly constructed dataset. (A1) When using the relaxation metrics such as binary metric or proportional metric, the F1 values of AE and OE have more than 10 points of improvement, which indicates that most of the detected terms overlap with the gold terms. For the PE task, the F1 scores are less than 50%, which is much different from 70% of the results on monolingual corpus in the previous works [62].

⁵<https://platform.openai.com/docs/models/gpt-3-5>

Table 4. F1 Scores of Bilingual Transfer from English to Chinese for AE and OE Tasks on Res (%)

Fusion Method	Model	Exact metric		Binary metric		Proportional metric	
		AE	OE	AE	OE	AE	OE
SAT	MBERT	65.43	48.68	82.44	72.72	76.49	63.53
	Adapter	65.27	50.54	81.43	73.67	77.17	64.57
	A-LaTran.	72.43	62.46	84.65	79.22	81.28	75.97
	M-LaTran.	73.53	64.38	85.56	80.84	81.96	77.20
SAF	MBERT	60.81	47.70	80.82	69.09	75.68	63.09
	Adapter	63.97	49.15	81.07	68.81	75.99	62.16
	A-LaTran.	72.10	61.68	82.58	77.83	80.70	74.03
	M-LaTran.	73.29	63.84	85.04	78.94	81.24	75.87
SF	MBERT	60.98	46.31	79.09	68.66	74.12	61.57
	Adapter	61.05	45.98	79.78	67.19	74.01	61.40
	A-LaTran.	69.89	61.16	82.46	77.46	79.14	73.68
	M-LaTran.	72.64	63.63	84.31	78.89	80.62	75.03
w/o sememe	MBERT	53.29	42.69	78.48	65.82	73.06	58.60
	Adapter	59.57	42.34	77.25	66.34	71.82	59.11
	A-LaTran.	69.73	59.53	82.16	75.09	77.22	71.22
	M-LaTran.	72.21	62.19	83.02	77.28	80.15	74.82
	ChatGPT	68.32	56.60	81.87	72.10	78.37	69.65
	MualTrn.	79.91	75.66	87.77	82.50	83.53	80.11

In addition, we also observed the following points: (1) Each model has better transfer performance in Res than in Lap, with an average of 13.06% and 10.36% higher in PE and TE, respectively. This may be due to the fact that there are more specific terms in the laptop domain. (2) On average, the performance on AE is about 14% higher than that on OE, which may be due to the greater diversity of opinion terms. (3) The F1 score of the AE task approaches the upper bound of this span-based model in monolingual settings (i.e., MualTrn.), indicating that transfer learning can indeed address aspect term extraction tasks in low-resource languages to some extent. (4) Compared to MualTrn., the model trained using the manually corrected training set, F1 values of the cross-lingual approaches are much lower, which further illustrates the extreme difficulty of zero-shot cross-lingual ABSA. (5) In all tasks, ChatGPT approaches the performance of the strong transfer methods, which is consistent with the previous assumption that ChatGPT has a strong cross-language capability. Utilizing ChatGPT for building datasets in low-resource languages, or distilling ChatGPT for a single language or task, are exciting directions.

6.2 Model Transfer vs. Label Transfer

As can be seen from Tables 4 and 5, label transfer has significantly better performance than model transfer, with an average of 9% and 7% higher for PE and TE, respectively. (A1) There are great differences in lexicon, syntax, semantics, and pragmatics between English and Chinese, which limits the performance of model transfer based on a pre-trained multilingual language model.

In model transfer, the performance of MBERT is very close to that of Adapter, with only about 1% difference in PE and TE, which confirms the prior established observations that the Adapter can achieve similar performance with only a few parameters optimized. In label transfer, M-LaTran. generally outperforms A-LaTran., and the possible reason for this is that the annotations in the

Table 5. Exact F1 Scores of Bilingual Transfer from English to Chinese for the PE and TE Tasks (%)

Fusion Method	Model	Res		Lap	
		PE	TE	PE	TE
SAT	MBERT	35.29	28.06	22.21	17.61
	Adapter	36.35	28.93	22.86	18.13
	A-LaTran.	43.74	34.54	27.91	22.17
	M-LaTran.	44.38	35.17	29.14	23.03
SAF	MBERT	32.96	26.34	21.33	16.94
	Adapter	33.97	26.95	21.41	16.96
	A-LaTran.	41.67	33.00	26.65	21.22
	M-LaTran.	42.19	33.93	27.46	22.66
SF	MBERT	30.29	23.97	19.45	15.48
	Adapter	31.04	24.65	20.71	16.47
	A-LaTran.	40.70	32.32	26.58	21.16
	M-LaTran.	41.67	33.41	27.24	22.42
w/o sememe	MBERT	27.63	22.14	15.24	12.20
	Adapter	27.98	22.23	16.05	12.83
	A-LaTran.	38.23	30.66	23.37	18.93
	M-LaTran.	39.60	31.91	24.68	20.11
	ChatGPT	36.48	34.13	22.67	18.12
	MualTrn.	66.14	61.81	50.97	48.95

training set obtained through M-LaTran. are less prone to boundary errors. As a result, the projected annotations contain less noise and have higher quality.

6.3 Results with the Sememe Bridge

Tables 4 and 5 demonstrate the effectiveness of our proposed sememe module, which significantly enhances both model transfer and label transfer. By introducing the sememe-encoded representation through the SAT fusion method, the F1 values for model transfer and label transfer in the PE task improved by 7.35% and 5.53%, respectively. (A2)

In terms of the fusion method, SAF performs better than SF. The performance is further improved by preserving the hierarchy of sense. (A3) We can summarize the following points: (1) The importance of the corresponding sememes of words varies in different contexts. Assigning weights to different sememes according to the context is a more efficient way to encode sememe information. (2) The structural information among word, sememe, and sense is beneficial for cross-lingual ABSA.

6.4 Impact of the Sememe Information

In the settings of cross-lingual learning, sememe representation was thoroughly trained and played a significant role in downstream tasks in the target language. However, the introduction of additional sememe information could also potentially impact the results of monolingual models. To evaluate the proportion of sememe information occupies in our method, we conduct an ablation study to quantitatively assess the contribution of sememe representation to the performance.

The experimental setup is as follows: In the comparative experiment, we use the backbone model without incorporating the sememe module. During prediction, sememe information is added in the form of an indicator vector. Specifically, we use categorical hashing to reduce the sememe

Table 6. The Most Frequently Occurring Sememes and Co-occurring Sememe Pairs on Res

	En^{trn}	Ch^{tst}_{manual}
Aspect	edible 食物, food 食品, human 人, material 材料, part 部件, mental 精神, fact 事情, eat 吃, place 地方, TakeCare 照料, location 位置, affairs 事务, help 帮助, flesh 肉, cook 烹调, time 时间, commerce 商业, put 放置	food 食品, edible 食物, part 部件, eat 吃, place 地方, human 人, affairs 事务, commerce 商业, TakeCare 照料, engage 从事, InstitutePlace 场所, vegetable 蔬菜, flesh 肉, organization 组织, space 空间, physical 物质
Opinion	BestQuality 最佳, able 能, urge 促使, fantastic 非常好, HighQuality 优质, FuncWord 功能词, FondOf 喜欢, entity 实体, big 大, strong 强, great 伟, grand 壮观, GreaterThanNormal 高于正常, important 重要	HighQuality 优质, FuncWord 功能词, FondOf 喜欢, very 很, like 爱惜, expression 词语, express 表示, able 能, urge 促使, advantageous 有利, intimate 亲密, BeWell 健壮, ExpressAgreement 示同意
POS	edible 食物-BestQuality 最佳, edible 食物-fantastic 非常好, material 材料-BestQuality 最佳, edible 食物-HighQuality 优质, mental 精神-BestQuality 最佳, edible 食物-FondOf 喜欢	food 食品-HighQuality 优质, edible 食物-HighQuality 优质, part 部件-HighQuality 优质, part 部件-FondOf 喜欢, food 食品-FondOf 喜欢, place 地方-HighQuality 优质
NEU	edible 食物-mediocre 可, edible 食物-ordinary 普通, mental 精神-mediocre 可, material 材料-mediocre 可, mental 精神-ordinary 普通, help 帮助-mediocre 可	edible 食物-FuncWord 功能词, food 食品-FuncWord 功能词, edible 食物-ordinary 普通, material 材料-FuncWord 功能词, edible 食物-ish 稍, part 部件-FuncWord 功能词
NEG	edible 食物-FuncWord 功能词, material 材料-FuncWord 功能词, mental 精神-FuncWord 功能词, fact 事情-inferior 劣质, TakeCare 照料-inferior 劣质, affairs 事务-inferior 劣质	edible 食物-neg 否, edible 食物-FuncWord 功能词, TakeCare 照料-inferior 劣, affairs 事务-inferior 劣质, engage 从事-inferior 劣质, TakeCare 照料-slow 慢

category to a numerical ID within 64, then it is converted into a 64-dimensional one-hot vector to stay consistent with other experimental settings. The sememe representation is then introduced using the parameter-free SF way. Note that this is not a monolingual setup, as we still use either the original training set in the source language or the pseudo-training set in the target language in the training stage.

Table 7 delineates that the direct incorporation of sememes exerts an inconsiderable influence on the experimental outcomes. Specifically, within the context of the MBERT model, there are observable increments of 1.83% for AE, 1.16% for OE, and 0.71%, 0.33% denominating PE and TE, correspondingly.

Nevertheless, it deserves to be highlighted that employing trained sememes catalyzes notably more substantial enhancements. On comparing the performances between trained sememe and w/o sememe, a marked increment of approximately 7.69% is evident for AE and 3.62%, 2.66%, 1.83% for OE, PE, and TE tasks, respectively, using the MBERT model.

These observations lead to an inference that while the straightforward integration of sememe information can elevate the performance of the models, sememe representation cultivated through the cross-lingual learning process can amplify the overall outcome more profoundly. In essence, it bolsters cross-lingual knowledge transfer from the source to the target language.

6.5 Rationality of the Sememe Bridge

To explore the rationality of the sememe bridge, we count the frequency of sememes corresponding to the terms in the Res dataset, as well as the number of co-occurrences of sememes corresponding to aspect and opinion terms in opinion pairs.

Table 6 shows some sememes with the highest frequency and the sememe pairs with the highest co-occurrence in the English training set and the Chinese test set. (A4) We find that: (1)

Table 7. Exact F1 Scores of Bilingual Transfer using SF Sememe Representation from English to Chinese on Res (%)

	Model	AE	OE	PE	TE
trained sememe	MBERT	60.98	46.31	30.29	23.97
	Adapter	61.05	45.98	31.04	24.65
	A-LaTran.	69.89	61.16	40.70	32.32
	M-LaTran.	72.64	63.63	41.67	33.41
untrained sememe	MBERT	55.12	43.85	28.34	22.47
	Adapter	59.54	42.42	28.37	22.42
	A-LaTran.	69.60	59.55	38.62	30.93
	M-LaTran.	72.29	62.24	39.74	32.02
w/o sememe	MBERT	53.29	42.69	27.63	22.14
	Adapter	59.57	42.34	27.98	22.23
	A-LaTran.	69.73	59.53	38.23	30.66
	M-LaTran.	72.21	62.19	39.60	31.91

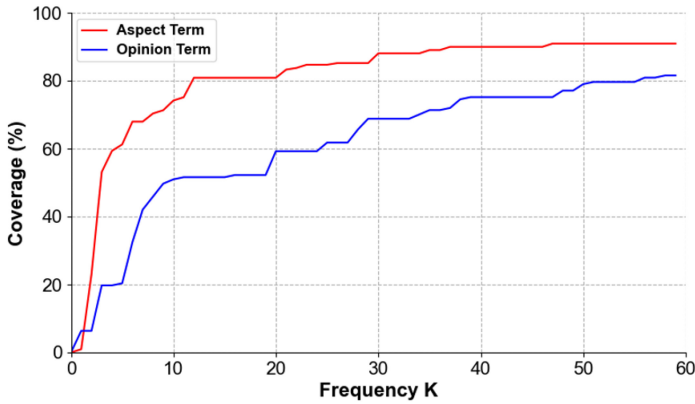


Fig. 11. Proportions w.r.t. different k values.

Sememes can provide obvious and reasonable information for the identification of aspect and opinion term detection, respectively. (2) There are some matching patterns between the sememe pairs. Capturing the relationship between sememe pairs can provide clues for term pairing. (3) Sememes corresponding to opinion terms is important for sentiment classification of opinion triplets. (4) There are a lot of overlaps between sememes and sememe pairs with high frequency in Chinese and English, which indicates that the sememe can act as an intermediate medium to bridge the gap between the source and the target language.

We further analyzed the missing terms of model transfer without sememe module but correctly predicted with SAT. (A4) Figure 11 shows the proportion of these missing terms' sememes in the highest frequency top K sememes. The figure shows that most of the missing terms can be covered by the top 50 sememes with the highest frequency, which can provide distinguishing features for ABSA.

6.6 Results of Zero-shot Cross-lingual Cross-domain Transfer

In order to investigate how the proposed sememe module performs on more difficult tasks, we conducted experiments under a cross-linguistic cross-domain setting. Table 8 shows the experimental

Table 8. Exact F1 Scores of Transfer Methods from English Res to Chinese Lap (%)

Fusion	Model	AE	OE	PE	TE
SAF	Adapter	56.51	46.99	26.28	20.33
	MBERT	56.43	47.68	26.14	20.02
	LaTran.	57.43	51.46	29.08	22.59
w/o sememe	Adapter	51.42	40.05	25.21	19.56
	MBERT	50.76	39.45	25.06	19.51
	LaTran.	52.51	45.99	27.85	21.66

results of transfer methods for English Res to Chinese Lap ABSA. We observe that: (1) The performances decrease after adding the cross-domain setting because of the further expansion of variety space between training and test set. (2) The gap between model transfer and label transfer has been narrowed. (3) Sememe knowledge can still improve performance because words in different fields still share a limited set of sememes.

7 Conclusion

In this work, we carry out the first study of Chinese OE, PE, and TE. First, we construct a parallel dataset of English and Chinese with the help of machine translation and a word aligner. Then, we investigate the effectiveness of cross-lingual learning on the newly constructed dataset, which can be used as benchmark dataset for further fantastic researchs. Third, we propose a plug-and-play sememe module, based on a semantic knowledge base, which connects Chinese and English bilingual languages through language-independent sememes. Experimental results indicate that the integration of semi-automatic data construction and transfer learning presents a viable solution for Chinese Aspect-Based Sentiment Analysis (ABSA). Furthermore, our innovative plug-and-play sememe module significantly enhances performance in both model transfer and label transfer. This proposed sememe module bears potential applicability to other cross-lingual tasks, including cross-lingual named entity recognition, event extraction, and semantic role labeling.

References

- [1] Wasi Uddin Ahmad, Zhisong Zhang, Xuezhe Ma, Kai-Wei Chang, and Nanyun Peng. 2019. Cross-lingual dependency parsing with unlabeled auxiliary languages. In *Proceedings of CoNLL*. 372–382.
- [2] Md. Shad Akhtar, Palaash Sawant, Sukanta Sen, Asif Ekbal, and Pushpak Bhattacharyya. 2019. Improving word embedding coverage in less-resourced languages through multi-linguality and cross-linguality: A case study with aspect-based sentiment analysis. *ACM Trans. Asian Low Resour. Lang. Inf. Process.* 18, 2 (2019), 15:1–15:22.
- [3] Massih-Reza Amini, Nicolas Usunier, and Cyril Goutte. 2009. Learning from multiple partially observed views - An application to multilingual text categorization. In *Proceedings of NeurIPS*. 28–36.
- [4] M. Saiful Bari, Shafiq R. Joty, and Prathyusha Jwalapuram. 2020. Zero-resource cross-lingual named entity recognition. In *Proceedings of AAAI*. 7415–7423.
- [5] Leonard Bloomfield. 1926. A set of postulates for the science of language. *Language* 2, 3 (1926), 153–164.
- [6] Jiahao Bu, Lei Ren, Shuang Zheng, Yang Yang, Jingang Wang, Fuzheng Zhang, and Wei Wu. 2021. ASAP: A Chinese review dataset towards aspect category sentiment analysis and rating prediction. In *Proceedings of NAACL-HLT*. 2069–2079.
- [7] Hongjie Cai, Rui Xia, and Jianfei Yu. 2021. Aspect-category-opinion-sentiment quadruple extraction with implicit aspects and opinions. In *Proceedings of ACL/IJCNLP*. 340–350.
- [8] Yitao Cai, Zhe Lin, and Xiaojun Wan. 2021. Making better use of bilingual information for cross-lingual AMR parsing. In *Findings of the ACL (Findings of ACL)*, Vol. ACL/IJCNLP 2021. 1537–1547.
- [9] Fang Chen and Yongfeng Huang. 2019. Knowledge-enhanced neural networks for sentiment analysis of Chinese reviews. *Neurocomputing* 368 (2019), 51–58.
- [10] Shaowei Chen, Jie Liu, Yu Wang, Wenzheng Zhang, and Ziming Chi. 2020. Synchronous double-channel recurrent network for aspect-opinion pair extraction. In *Proceedings of ACL*. 6515–6524.

- [11] Shaowei Chen, Yu Wang, Jie Liu, and Yuelin Wang. 2021. Bidirectional machine reading comprehension for aspect sentiment triplet extraction. In *Proceedings of AAAI*. 12666–12674.
- [12] Yang Chen, Chao Jiang, Alan Ritter, and Wei Xu. 2023. Frustratingly easy label projection for cross-lingual transfer. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Findings)*.
- [13] Zhuang Chen and Tiejun Qian. 2020. Enhancing aspect term extraction with soft prototypes. In *Proceedings of EMNLP*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). 2107–2117.
- [14] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*. 4171–4186.
- [15] Zhendong Dong and Qiang Dong. 2003. HowNet-a hybrid language and knowledge resource. In *International Conference on Natural Language Processing and Knowledge Engineering, 2003. Proceedings. 2003. IEEE*, 820–824.
- [16] Xiangyu Duan, Jun Zhao, and Bo Xu. 2007. Word sense disambiguation through sememe labeling. In *Proceedings of IJCAI*. 1594–1599.
- [17] Chris Dyer, Victor Chahuneau, and Noah A. Smith. 2013. A simple, fast, and effective reparameterization of IBM Model 2. In *Proceedings of NAACL*. 644–648.
- [18] Hongliang Fei and Ping Li. 2020. Cross-lingual unsupervised sentiment classification with multi-view transfer learning. In *Proceedings of ACL*. 5759–5771.
- [19] Hao Fei, Meishan Zhang, and Donghong Ji. 2020. Cross-lingual semantic role labeling with high-quality translated training corpus. In *Proceedings of ACL*. 7014–7026.
- [20] Hao Fei, Meishan Zhang, Fei Li, and Donghong Ji. 2020. Cross-lingual semantic role labeling with model transfer. *IEEE ACM Trans. Audio Speech Lang. Process.* 28 (2020), 2427–2437.
- [21] Bairu Hou, Fanchao Qi, Yuan Zang, Xurui Zhang, Zhiyuan Liu, and Maosong Sun. 2020. Try to substitute: An unsupervised Chinese word sense disambiguation method based on HowNet. In *Proceedings of COLING*. 1752–1757.
- [22] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *Proceedings of ICML*, Vol. 97. 2790–2799.
- [23] Karthikeyan K., Zihan Wang, Stephen Mayhew, and Dan Roth. 2020. Cross-lingual ability of multilingual BERT: An empirical study. In *Proceedings of ICLR*.
- [24] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In *NeurIPS*.
- [25] Guillaume Lample, Alexis Conneau, Marc’Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. 2018. Word translation without parallel data. In *Proceedings of ICLR*.
- [26] Peifeng Li and Guodong Zhou. 2012. Employing morphological structures and sememes for Chinese event extraction. In *Proceedings of COLING*. 1619–1634.
- [27] Xin Li, Lidong Bing, Piji Li, and Wai Lam. 2019. A unified model for opinion target extraction and target sentiment prediction. In *Proceedings of AAAI*. 6714–6721.
- [28] Ziran Li, Ning Ding, Zhiyuan Liu, Haitao Zheng, and Ying Shen. 2019. Chinese relation extraction with multi-grained information and external linguistic knowledge. In *Proceedings of ACL*. 4377–4386.
- [29] Jian Liao, Min Wang, Xin Chen, Suge Wang, and Kai Zhang. 2022. Dynamic commonsense knowledge fused method for Chinese implicit sentiment analysis. *Inf. Process. Manag.* 59, 3 (2022), 102934.
- [30] Bing Liu. 2012. *Sentiment Analysis and Opinion Mining*. Morgan & Claypool Publishers.
- [31] Lizhen Liu, Mengyun Lei, and Hanshi Wang. 2013. Combining domain-specific sentiment lexicon with HowNet for Chinese sentiment analysis. *J. Comput.* 8, 4 (2013), 878–883.
- [32] Pengyuan Liu, Yongsheng Tian, Chengyu Du, and Likun Qiu. 2020. Construction and analysis of Chinese multi-target sentiment classification dataset. In *Proceedings of the 19th Chinese National Conference on Computational Linguistics*. 604–615.
- [33] Chenwei Lou, Jun Gao, Changlong Yu, Wei Wang, Huan Zhao, Weiwei Tu, and Ruifeng Xu. 2022. Translation-based implicit annotation projection for zero-shot cross-lingual event argument extraction. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR’22)*. Association for Computing Machinery, 2076–2081.
- [34] Ling Luo, Xiang Ao, Yan Song, Jinyao Li, Xiaopeng Yang, Qing He, and Dong Yu. 2019. Unsupervised neural aspect extraction with sememes. In *Proceedings of IJCAI*. ijcai.org, 5123–5129.
- [35] Ruixuan Luo, Jingjing Xu, Yi Zhang, Xuancheng Ren, and Xu Sun. 2019. PKUSEG: A toolkit for multi-domain Chinese word segmentation. *CoRR* abs/1906.11455 (2019).
- [36] Dehong Ma, Sujian Li, Fangzhao Wu, Xing Xie, and Houfeng Wang. 2019. Exploring sequence-to-sequence learning in aspect term extraction. In *Proceedings of ACL*. 3538–3547.
- [37] Yilin Niu, Ruobing Xie, Zhiyuan Liu, and Maosong Sun. 2017. Improved word representation learning with sememes. In *Proceedings of ACL*. 2049–2058.

- [38] Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Yoav Goldberg, Jan Hajic, Christopher D. Manning, Ryan T. McDonald, Slav Petrov, Sampo Pyysalo, Natalia Silveira, Reut Tsarfaty, and Daniel Zeman. 2016. Universal dependencies v1: A multilingual treebank collection. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation LREC 2016, Portorož, Slovenia, May 23–28, 2016*.
- [39] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *NeurIPS*.
- [40] Haiyun Peng, Lu Xu, Lidong Bing, Fei Huang, Wei Lu, and Luo Si. 2020. Knowing what, how and why: A near complete solution for aspect-based sentiment analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*. 8600–8607.
- [41] Matúš Pikuliak, Marián Simko, and Mária Bielíková. 2021. Cross-lingual learning for text processing: A survey. *Expert Syst. Appl.* 165 (2021), 113765.
- [42] Telmo Pires, Eva Schlinger, and Dan Garrette. 2019. How multilingual is multilingual BERT?. In *Proceedings of ACL*. 4996–5001.
- [43] Barbara Plank. 2021. Cross-lingual cross-domain nested named entity evaluation on English web texts. In *Proceedings of Findings of ACL (Findings of ACL)*, Vol. ACL/IJCNLP 2021. 1808–1815.
- [44] Maria Pontiki, Dimitris Galanis, Haris Papageorgiou, Ion Androutsopoulos, Suresh Manandhar, Mohammad Al-Smadi, Mahmoud Al-Ayyoub, Yanyan Zhao, Bing Qin, Orphée De Clercq, Véronique Hoste, Marianna Apidianaki, Xavier Tannier, Natalia V. Loukachevitch, Evgeniy V. Kotelnikov, Núria Bel, Salud María Jiménez-Zafra, and Gülsen Eryigit. 2016. SemEval-2016 task 5: Aspect based sentiment analysis. In *International Workshop on Semantic Evaluation*. 19–30.
- [45] Maria Pontiki, Dimitris Galanis, Harris Papageorgiou, Suresh Manandhar, and Ion Androutsopoulos. 2015. SemEval-2015 task 12: Aspect based sentiment analysis. In *Proceedings of the 9th International Workshop on Semantic Evaluation (SemEval’15)*. 486–495.
- [46] Maria Pontiki, Dimitris Galanis, John Pavlopoulos, Harris Papageorgiou, Ion Androutsopoulos, and Suresh Manandhar. 2014. SemEval-2014 task 4: Aspect based sentiment analysis. In *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval’14)*. 27–35.
- [47] Peter Prettenhofer and Benno Stein. 2010. Cross-language text classification using structural correspondence learning. In *Proceedings of ACL*. 1118–1127.
- [48] Fanchao Qi, Liang Chang, Maosong Sun, Sicong Ouyang, and Zhiyuan Liu. 2020. Towards building a multilingual sememe knowledge base: Predicting sememes for BabelNet synsets. In *Proceedings of AAAI*. 8624–8631.
- [49] Fanchao Qi, Chenghao Yang, Zhiyuan Liu, Qiang Dong, Maosong Sun, and Zhendong Dong. 2019. OpenHowNet: An open sememe-based lexical knowledge base. *arXiv preprint arXiv:1901.09957* (2019).
- [50] Alan Ramponi and Barbara Plank. 2020. Neural unsupervised domain adaptation in NLP - A survey. In *Proceedings of COLING*. 6838–6855.
- [51] Erik F. Tjong Kim Sang. 2002. Introduction to the CoNLL-2002 shared task: Language-independent named entity recognition. In *Proceedings of CoNLL*, Dan Roth and Antal van den Bosch (Eds.).
- [52] Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of CoNLL*. 142–147.
- [53] Nitish Srivastava, Geoffrey E. Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: A simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15, 1 (2014), 1929–1958.
- [54] Stephanie M. Strassel and Jennifer Tracey. 2016. LORELEI language packs: Data, tools, and resources for technology development in low resource languages. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation LREC 2016, Portorož, Slovenia, May 23–28, 2016*.
- [55] Chi Sun, Luyao Huang, and Xipeng Qiu. 2019. Utilizing BERT for aspect-based sentiment analysis via constructing auxiliary sentence. In *Proceedings of NAACL-HLT*. 380–385.
- [56] Jörg Tiedemann, Zeljko Agic, and Joakim Nivre. 2014. Treebank translation for cross-lingual parser induction. In *Proceedings of CoNLL*. 130–140.
- [57] Deqing Wang, Baoyu Jing, Chenwei Lu, Junjie Wu, Guannan Liu, Chenguang Du, and Fuzhen Zhuang. 2021. Coarse alignment of topic and sentiment: A unified model for cross-lingual sentiment classification. *IEEE Trans. Neural Networks Learn. Syst.* 32, 2 (2021), 736–747.
- [58] Fang Wang, Yuncong Li, Wenjun Zhang, and Sheng-hua Zhong. 2021. A more fine-grained aspect-sentiment-opinion triplet extraction task. *CoRR* abs/2103.15255 (2021). [arXiv:2103.15255](https://arxiv.org/abs/2103.15255)
- [59] Wenya Wang and Sinno Jialin Pan. 2018. Recursive neural structural correspondence network for cross-domain aspect and opinion co-extraction. In *Proceedings of ACL*. 2171–2181.
- [60] Wenya Wang and Sinno Jialin Pan. 2018. Transition-based adversarial network for cross-lingual aspect extraction. In *Proceedings of IJCAI*. ijcai.org, 4475–4481.

- [61] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP - Demos*. 38–45.
- [62] Zhen Wu, Chengcan Ying, Fei Zhao, Zhifang Fan, Xinyu Dai, and Rui Xia. 2020. Grid tagging scheme for end-to-end fine-grained opinion extraction. In *Findings of EMNLP*. 2576–2585.
- [63] Qingrong Xia, Bo Zhang, Rui Wang, Zhenghua Li, Yue Zhang, Fei Huang, Luo Si, and Min Zhang. 2021. A unified span-based approach for opinion mining with syntactic constituents. In *Proceedings of NAACL-HLT*. 1795–1804.
- [64] Jingjing Xu, Hao Zhou, Chun Gan, Zaixiang Zheng, and Lei Li. 2021. Vocabulary learning via optimal transport for neural machine translation. In *Proceedings of ACL/IJCNLP*. 7361–7373.
- [65] Kang Xu, Fei Li, Dongdong Xie, and Donghong Ji. 2022. Revisiting aspect-sentiment-opinion triplet extraction: Detailed analyses towards a simple and effective span-based model. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 30 (2022), 2918–2927.
- [66] Xiaoying Xu, Ya Li, Liping Hu, and Jianhua Tao. 2009. Categorizing terms' subjectivity and polarity manually for opinion mining in Chinese. In *2009 3rd International Conference on Affective Computing and Intelligent Interaction and Workshops*. 1–6.
- [67] Heng Yang, Biqing Zeng, Jianhao Yang, Youwei Song, and Ruyang Xu. 2021. A multi-task learning model for Chinese-oriented aspect polarity classification and aspect term extraction. *Neurocomputing* 419 (2021), 344–356.
- [68] Yichun Yin, Furu Wei, Li Dong, Kaimeng Xu, Ming Zhang, and Ming Zhou. 2016. Unsupervised word and dependency path embeddings for aspect term extraction. In *Proceedings of IJCAI*. 2979–2985.
- [69] Chen Zhang, Qiuchi Li, Dawei Song, and Benyou Wang. 2020. A multi-task learning framework for opinion triplet extraction. In *Findings of EMNLP*, Trevor Cohn, Yulan He, and Yang Liu (Eds.). 819–828.
- [70] Meishan Zhang, Yue Zhang, and Guohong Fu. 2019. Cross-lingual dependency parsing using code-mixed TreeBank. In *Proceedings of EMNLP-IJCNLP*. 997–1006.
- [71] Wenxuan Zhang, Ruidan He, Haiyun Peng, Lidong Bing, and Wai Lam. 2021. Cross-lingual aspect-based sentiment analysis with aspect term code-switching. In *Proceedings of EMNLP*. 9220–9230.
- [72] He Zhao, Longtao Huang, Rong Zhang, Quan Lu, and Hui Xue. 2020. SpanMlt: A span-based multi-task learning framework for pair-wise aspect and opinion terms extraction. In *Proceedings of ACL*. 3239–3248.
- [73] Jun Zhao, Hongbo Xu, Xuanjing Huang, Songbo Tan, and Qi Zhang. 2008. Overview of Chinese opinion analysis evaluation 2008. (2008).
- [74] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2023. A survey of large language models. *CoRR abs/2303.18223* (2023). arXiv:2303.18223
- [75] Ranran Zhen, Rui Wang, Guohong Fu, Chengguo Lv, and Meishan Zhang. 2021. Chinese opinion role labeling with corpus translation: A pivot study. In *Proceedings of EMNLP*. 10139–10149.

Received 26 August 2022; revised 8 June 2024; accepted 12 August 2024